



PreLab 05 – Worksheet

Name: Mohammad Husain Naqvi (11062) Syed Fakhar Iqbal Rizvi	ID: 11062 11091	Section: T-4
--	--------------------	--------------

Note: Assumptions and logics should be explained separately in tasks after the task results.

Task 1

Exercise

Exercise 1: Identify Synthesizability

```
always @(*) begin
    for (i = 0; i < n; i = i + 1)
        sum = sum + a[i];
end
```

No, the above code is not synthesizable due to the following reasons:

1. 'sum' not being initialized in the loop.
2. The variable n is also not provided with any value or constant.

Corrected code:

```
parameter n = 4;
integer i;

always @(*) begin
    sum = 0; // Initialize to prevent latch inference
    for (i = 0; i < n; i = i + 1) begin
        sum = sum + a[i];
    end
end
```



Task 2

Exercise

Exercise 2: Fix the Code

```
always begin
    #10 clk = ~clk;
end
```

Fixed Code:

```
initial begin
    clk = 0; // Initialize to define the starting state
end

always begin
    #10 clk = ~clk; // Period = 20 time units (50 MHz if 1 ns timescale)
end
```

Task 3

Exercise

Exercise 3: What hardware does this code describe?

```
always @(posedge clk) begin
    if (en)
        q <= d;
```

```
end
```

This code basically describes a positive edge triggered D flip flop. With an enable variable 'en'.

q and d have a non-blocking assignment between them.

- When en = 1 and clock is on positive edge. q will get the value of d.
- When en = 0 and clock is on positive edge. q will not get the value of d and remain in their current state.